Introduction to Gen AI & Agentic AI



- Introduction to Generative AI
- LLM's
- What is Agentic AI
- Agentic AI vs Generative AI
- Applications of Agentc AI

Presented by - G D Abhishek, AI Developer

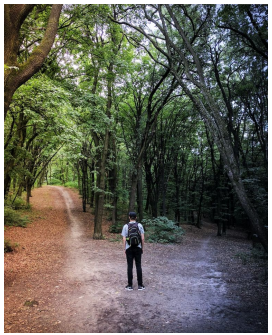
Artificial Intelligence

Artificial Intelligence

What is Intelligence?

Are we intelligent?

1 Decision Making

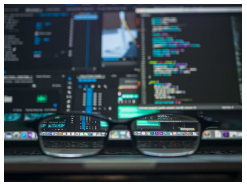




2 Recognition







Artificial Intelligence

Are we intelligent?

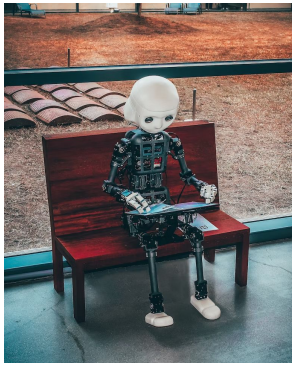
3 Communication





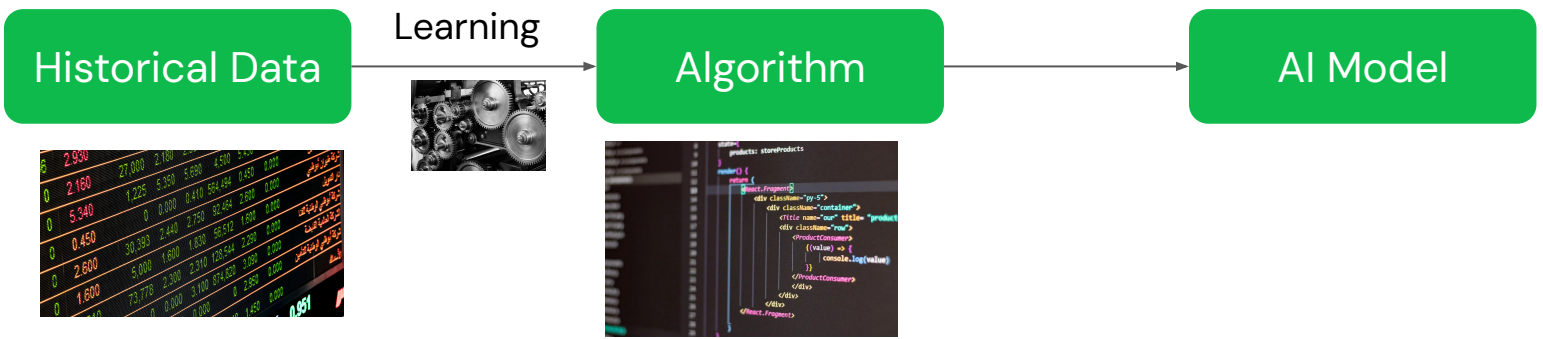
Artificial Intelligence

Artificial Intelligence (AI) refers to the ability of machines or computer systems to perform tasks that would typically require human intelligence, such as understanding natural language, recognizing images, making decisions, and solving problems



Artificial Intelligence

AI is achieved through the use of algorithms and statistical models that allow machines to learn from data, adapt to new situations, and improve their performance over time. The ultimate goal of AI is to create intelligent machines that can perform complex tasks and help us solve real-world problems.



Data ?

Data refers to any information or set of facts that can be stored, processed, or analyzed by a computer system

1

Tabular Data

	A	B	C	D	E	F
1	Country	Salesperson	Order Date	OrderID	Units	Order Amount
2	USA	Fuller	1/01/2011	10392	13	1,440.00
3	UK	Gloucester	2/01/2011	10397	17	716.72
4	UK	Bromley	2/01/2011	10771	18	344.00
5	USA	Finchley	3/01/2011	10393	16	2,556.95
6	USA	Finchley	3/01/2011	10394	10	442.00
7	UK	Gillingham	3/01/2011	10395	9	2,122.92
8	USA	Finchley	6/01/2011	10396	7	1,903.80
9	USA	Callahan	8/01/2011	10399	17	1,765.60
10	USA	Fuller	8/01/2011	10404	7	1,591.25
11	USA	Fuller	9/01/2011	10398	11	2,505.60
12	USA	Coghill	9/01/2011	10403	18	855.01
13	USA	Finchley	10/01/2011	10401	7	3,868.60
14	USA	Callahan	10/01/2011	10402	11	2,713.50
15	UK	Rayleigh	13/01/2011	10406	15	1,830.78
16	USA	Callahan	14/01/2011	10408	10	1,622.40
17	USA	Farnham	14/01/2011	10409	19	319.20
18	USA	Farnham	15/01/2011	10410	16	802.00

2

Image or Video



3

Text

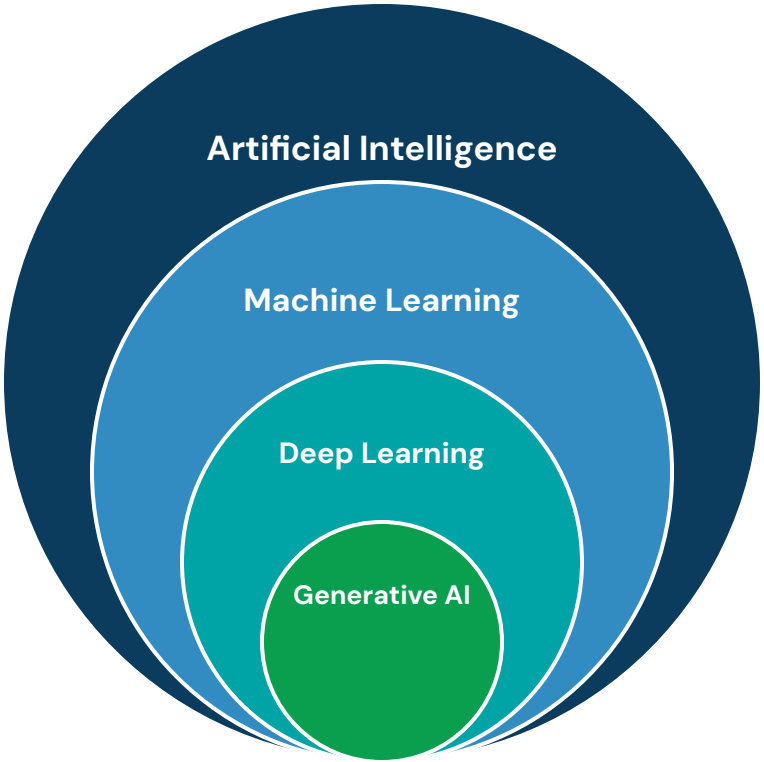


4

Audio



AI Branches



Artificial Intelligence

Systems that mimic human intelligence to solve complex tasks and adapt to new inputs.

Machine Learning

A subset of AI using statistical models to learn and make predictions from historical data.

Deep Learning

Multi-layered neural networks trained to extract high-level features from unstructured data.

Generative AI

An advanced branch that creates completely new data—like text, images, or audio.

Generative AI

Generative AI is a branch of artificial intelligence that generates data, like text, images, or music, based on patterns learned from data. It's widely used in creative tasks, automation, and content creation.

01

Historical Data

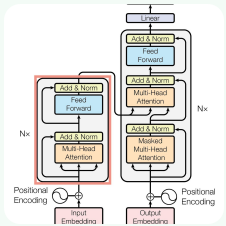


02

Learning Process

Transformers / GAN's etc.





03

GenAI Model

Task Action

↓

Generate Content

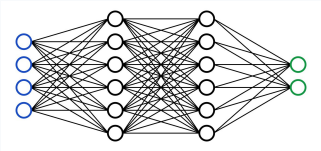
Traditional AI vs. Generative AI

Traditional AI – ML/DL (Predictive Pipeline)

Historical Data



Algorithm
(Learning Process)



ML/DL Model

Trained on patterns to classify or regression

Prediction

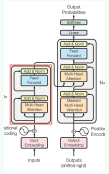
Outputs specific decision or numerical value

Generative AI (Creation Pipeline)

Historical Data



Transformers / GANs
(Deep Architectures)

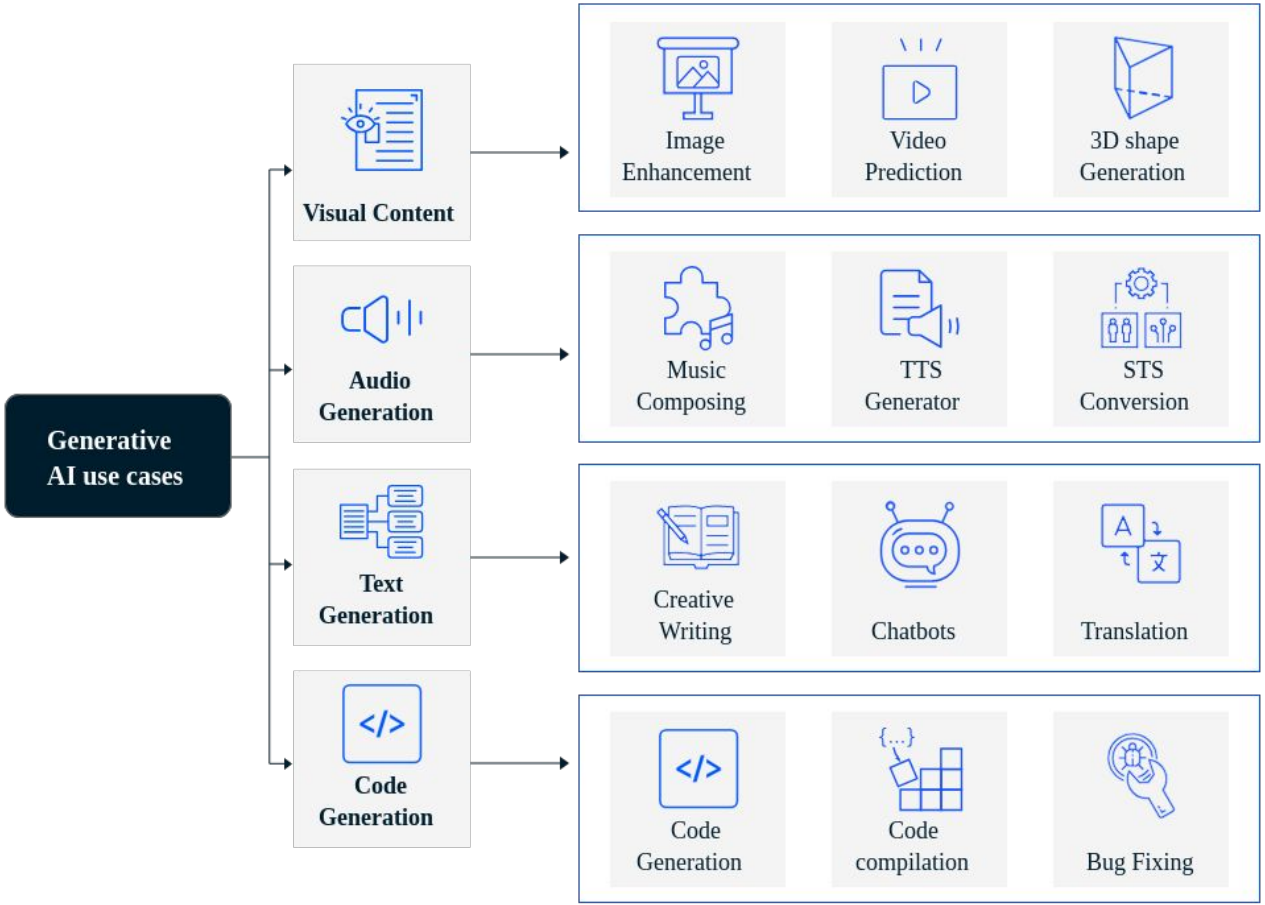


GenAI Model

Learns underlying distribution of data

Generate Content

Creates brand new, original data outputs

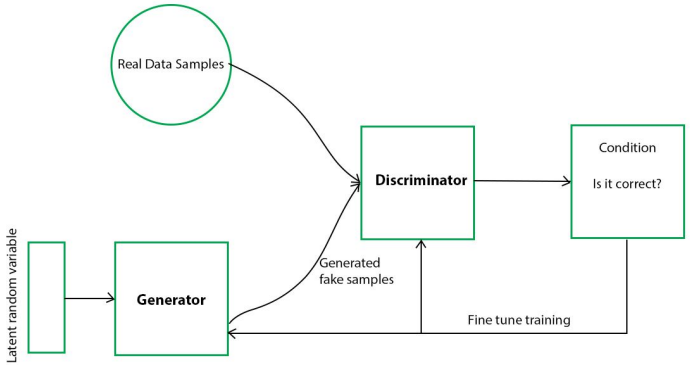

Generative AI use cases
Visual Content
Audio Generation
Text Generation
Code Generation
Image Enhancement
Video Prediction
3D shape Generation
Music Composing
TTS Generator
STS Conversion
Creative Writing
Chatbots
Translation
Code Generation
Code compilation
Bug Fixing

Rise of Generative AI

GAN's (2014):

Proposed by Ian Goodfellow, GANs introduced a new way to generate realistic images, videos, and data by pitting two neural networks (a generator and a discriminator) against each other.

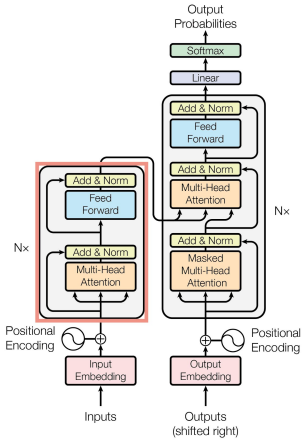
- This was the first time AI could generate high-quality, realistic images that were indistinguishable from real photos, sparking major interest in generative AI.

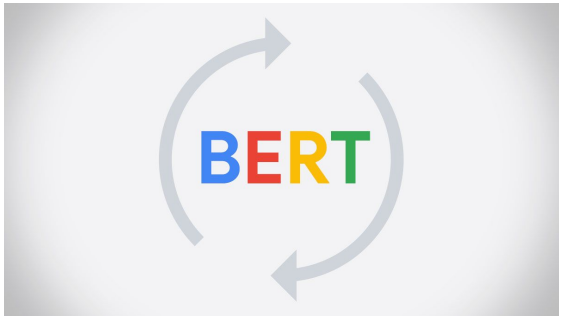


Rise of Generative AI

Transformers (2017):

- Introduced in the paper "Attention is All You Need," transformers revolutionized natural language processing by allowing models to handle sequential data (like text) with more efficiency and power than before.
- Transformers are the backbone of advanced language models, enabling complex text generation and understanding.




BERT


Fine-
tune
T5


GPT
Generative Pre-trained Transformer

Potential Generative AI Use Cases

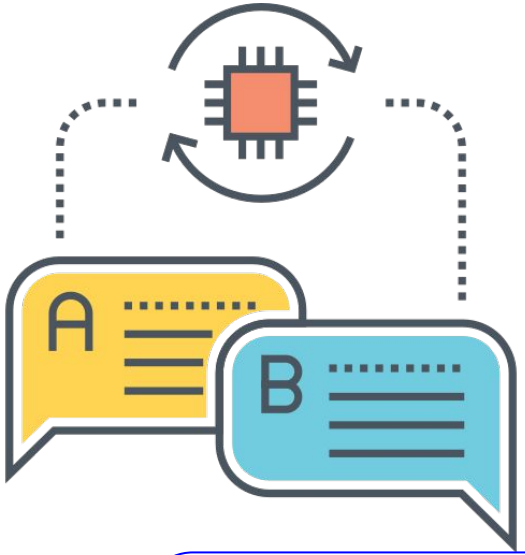


LLM’s



What is LLM?

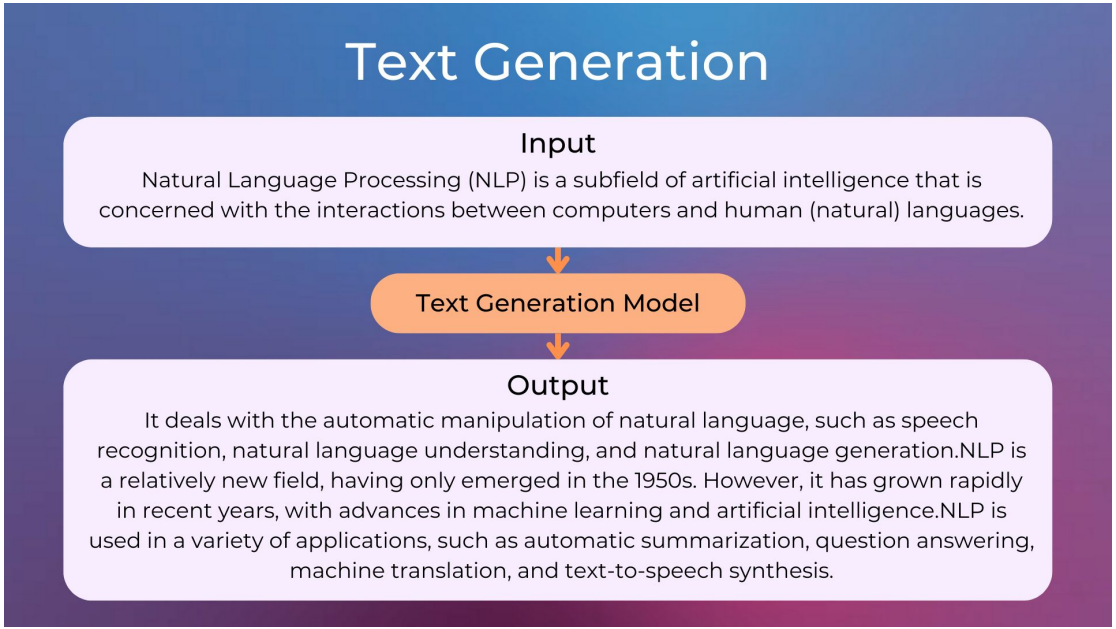
LLM stands for **"Large Language Model,"** which refers to a type of machine learning model that is trained on a **large amount of text data** to u**nderstand and generate human-like text**.



LLMs are often used in **natural language processing (NLP)** tasks such as **language translation, text summarization, and text generation**. Examples of LLMs include OpenAI's GPT, llama2, Gemini etc...

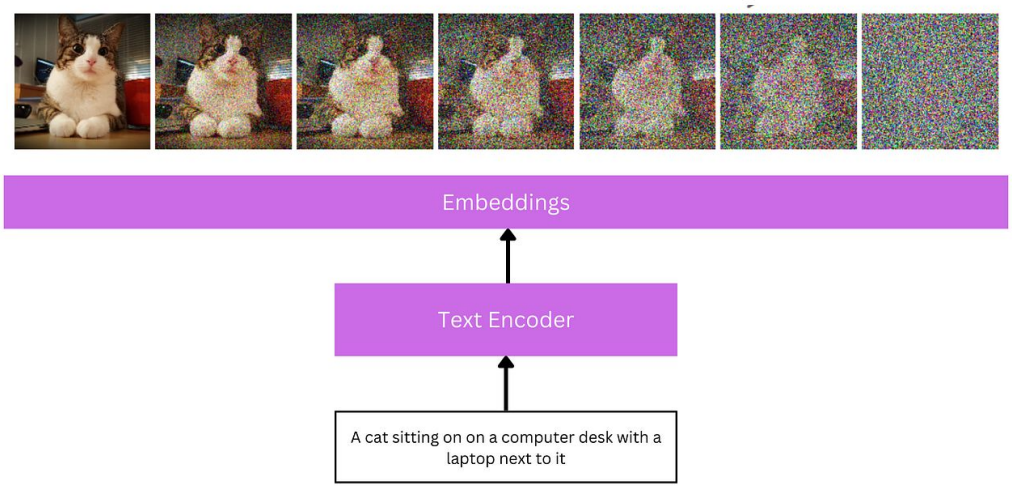
Types of Generative AI Models

Text to text model



Types of Generative AI Models

Text to Image models



https://colab.research.google.com/github/stability-ai/stability-sdk/blob/main/nbs/Stable_Image_API_Public.ipynb#scrollTo=AvTo6cVeF3ip

Types of Generative AI Models

Text to Video models

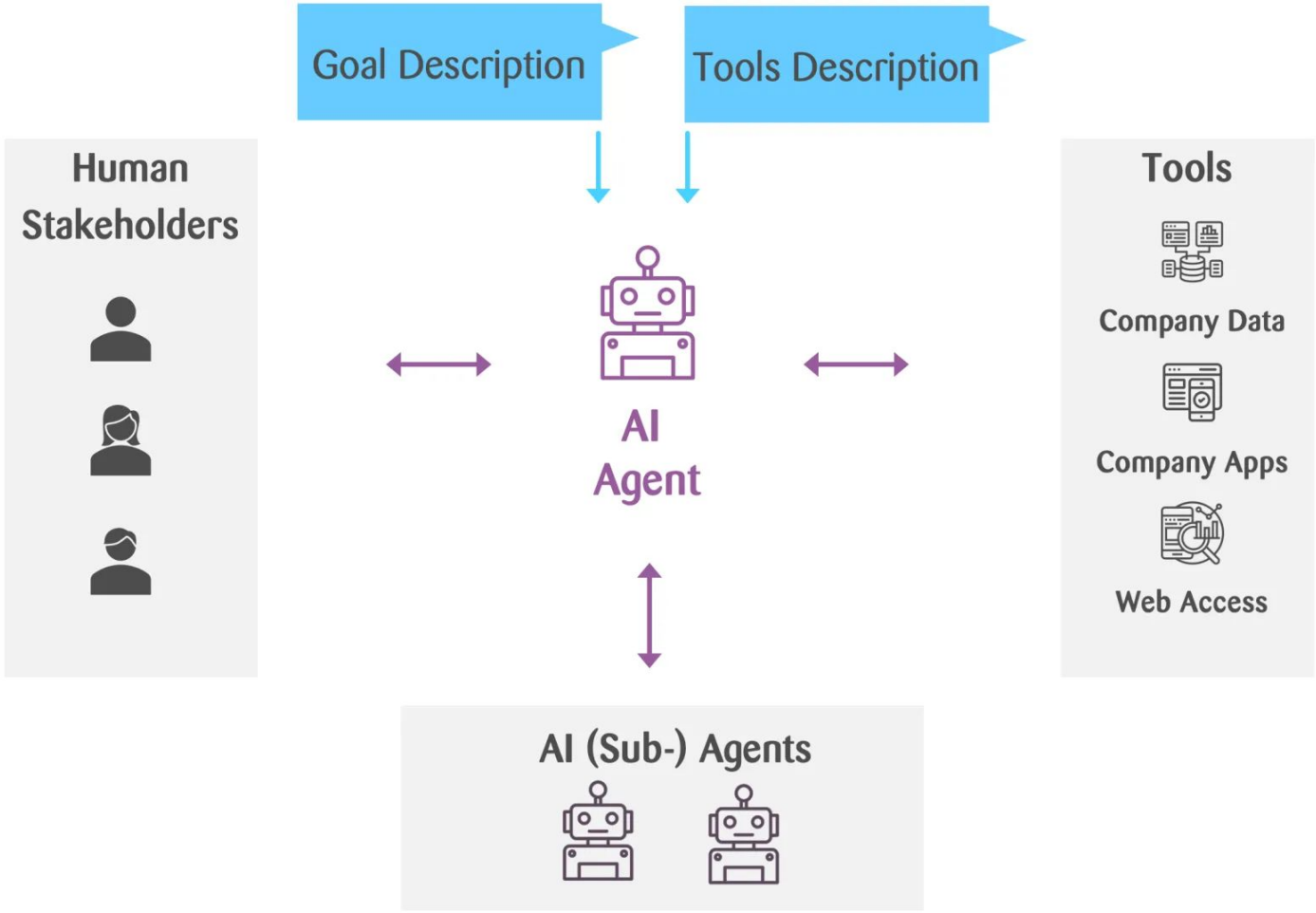
A person jogging through a lush green park on a sunny day, surrounded by tall trees, with a peaceful path winding through the park.

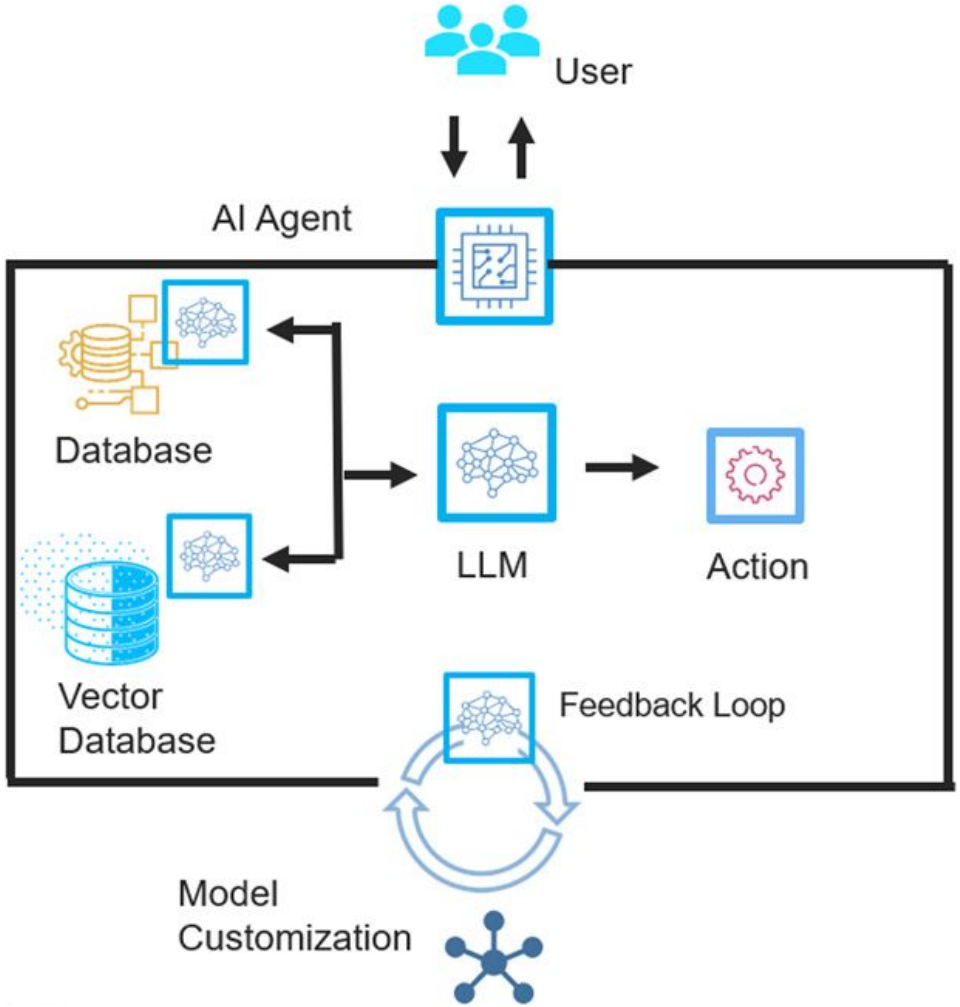
LLM



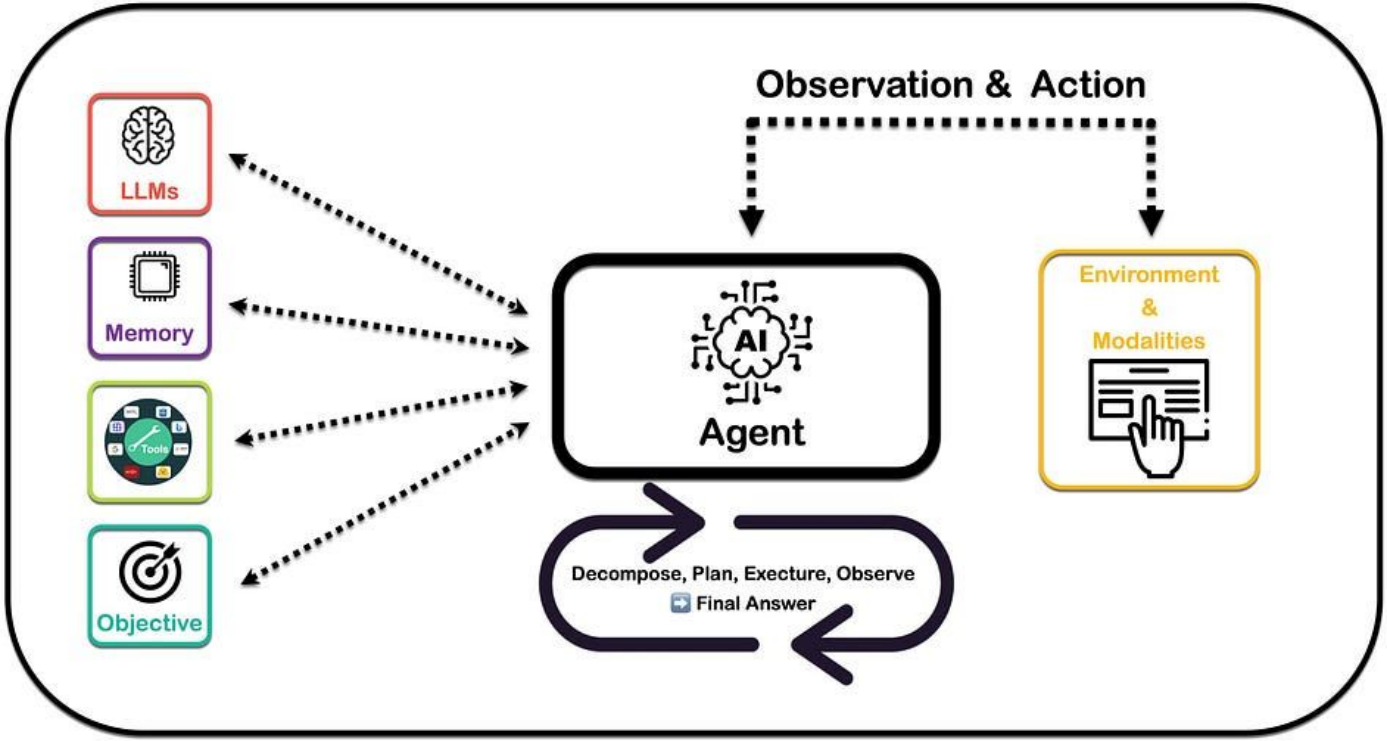
https://phenaki.video/
https://makeavideo.studio/

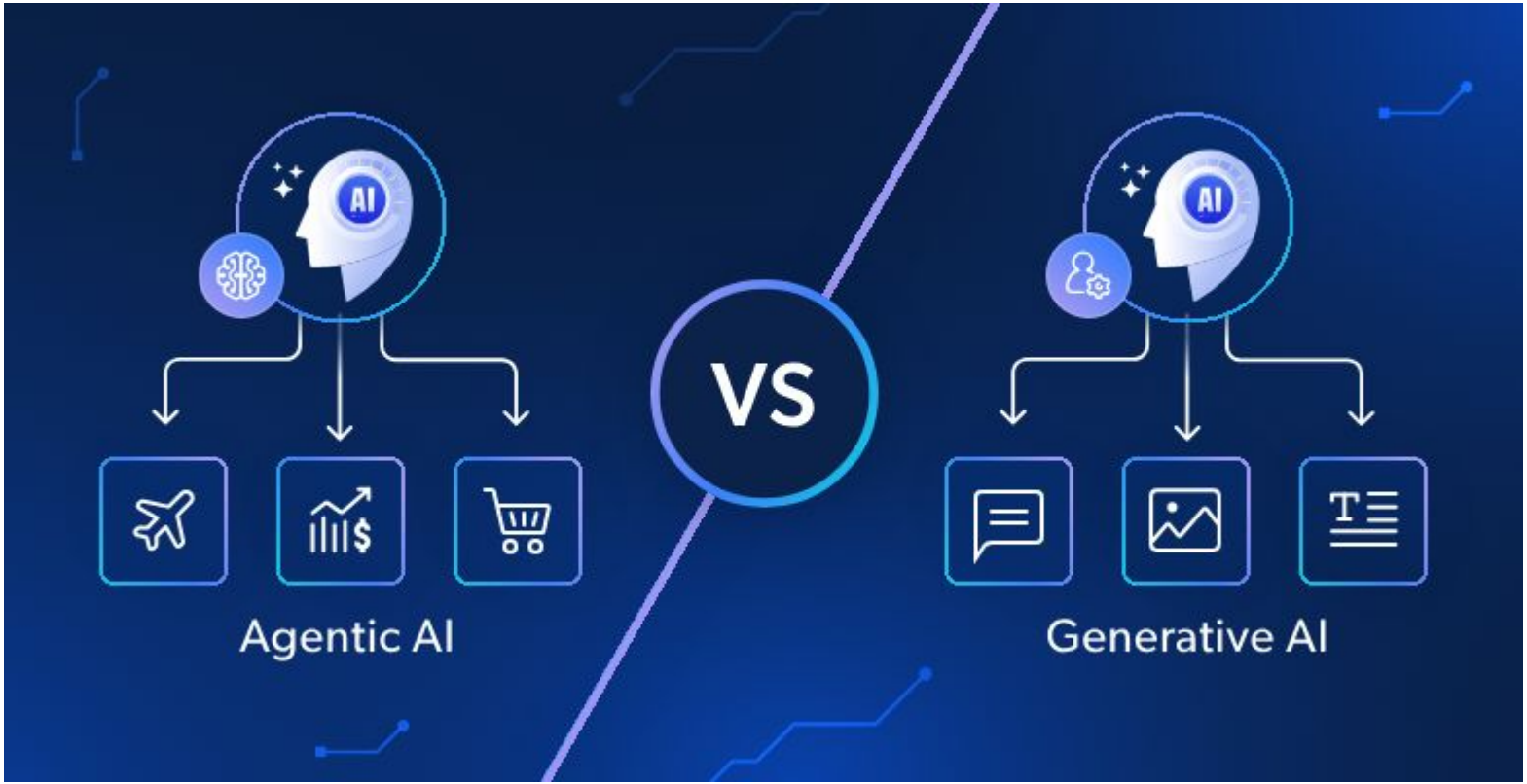
Agentic AI


Goal Description
Tools Description
Human Stakeholders
AI Agent
Tools
Company Data
Company Apps
Web Access
AI (Sub-) Agents

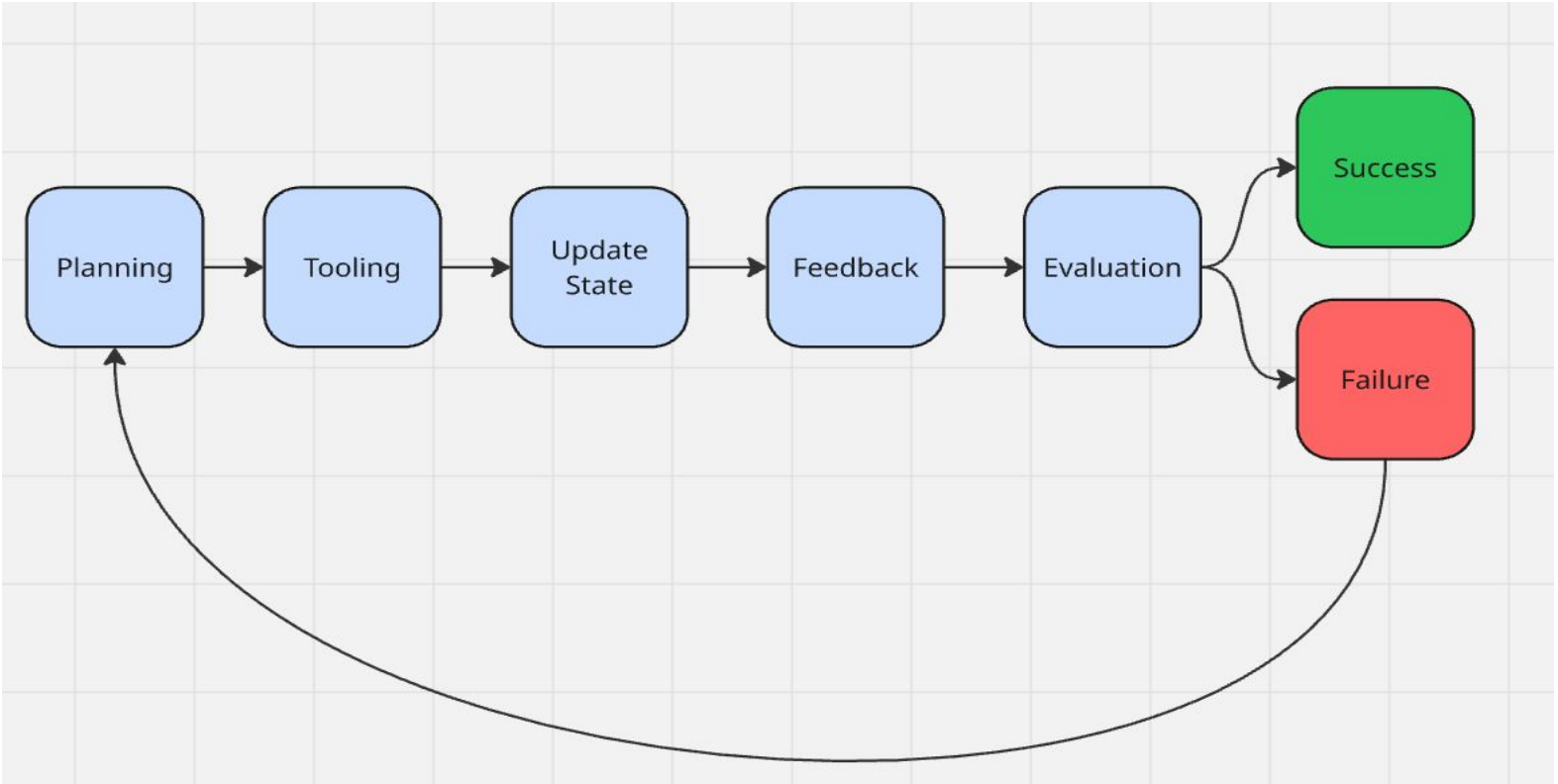

User
AI Agent
Database
Vector
Database
LLM
Action
Feedback Loop
Model
Customization

AI Agents / Agentic Applications




AI
VS
AI
Agentic AI
Generative AI

Criteria	Agentic AI	Generative AI
Primary Purpose	✔️ Task execution and completion	✔️ Content generation and synthesis
Goal-Oriented	✔️ Yes — operates with defined objectives	❌ Not inherently goal-driven
Prompt Dependency	❌ Minimal — often runs autonomously	✔️ High — prompt is required to initiate output
Output Type	✔️ State change, completed actions	✔️ Text, images, code, structured data
Tool Use / API Access	✔️ Frequently uses tools and functions	❌ Rarely (unless wrapped in agent loop)
Memory Requirements	✔️ Needs short- and long-term memory	❌ Optional (only in RAG or custom builds)
Control Flow	✔️ Feedback loop with retry logic	❌ One-shot generation (no loop by default)
Autonomy	✔️ High autonomy possible	❌ Human-in-the-loop is typical
Evaluation Method	✔️ Binary success/failure	✔️ Subjective quality (originality, tone, etc.)
Real-World Examples	✔️ Web scraping bots, self-driving cars	✔️ ChatGPT, DALL·E, GitHub Copilot


Planning
Tooling
Update State
Feedback
Evaluation
Success
Failure